Department of Artificial Intelligence, SVNIT,SURAT

B.Tech-III ,SEM-V

Subject- Machine Learning(AI301)

LAB ASSIGNMENT-10

Problem statement:

a. Develop and train an Artificial Neural Network (ANN) to classify handwritten digits (0-9) using the MNIST dataset.
b. Load and preprocess the MNIST dataset (normalize images and split into train/test sets).
c. Implement an ANN with one input layer, hidden layers, and a softmax output layer.
d. Train the model using cross-entropy loss and a suitable optimizer (e.g., Adam).
e. Evaluate the model using accuracy, precision, and recall on the test set.
f. Compare the performance with standard benchmarks and analyze the impact of hyperparameters.
g. Dataset: MNIST dataset contains 70,000 grayscale images of digits. The dataset is publicly available and can be accessed via frameworks like TensorFlow, PyTorch, or Keras.
h. Or Link: https://git-disl.github.io/GTDLBench/datasets/mnist_datasets/